

Coin Diameter Measurement Using OpenCV and AI

Web-based system for measuring real-world coin dimensions using Python, OpenCV, and computer vision techniques.



Project Overview

Input
User uploads image of coins via web interface

Processing
OpenCV detects coins and calculates pixel-to-millimeter conversion

Output
Real-world diameter measurements in millimeters with visual overlay

This application bridges digital images with physical measurements using computer vision fundamentals.

Getting Started





Technologies Stack

- Python for core logic
- OpenCV for image processing
- NumPy for mathematical operations
- Flask for web interface
- HTML for frontend

Core Objectives

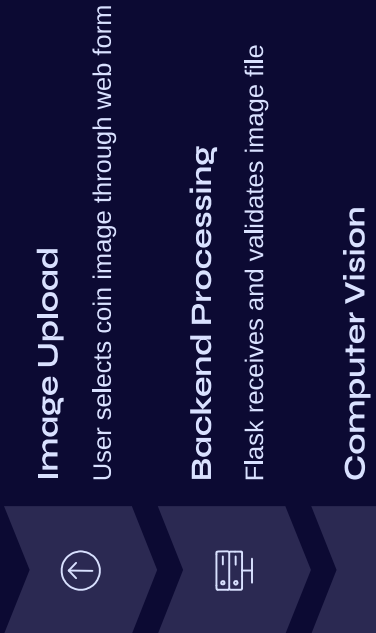
o1	o2	o3
Detect Multiple Coins or shapes Identify circular objects from uploaded images using contour detection and circularity filtering	Real-World Measurement Calculate actual diameter in millimeters using pixel-to-millimeter conversion with reference shape	Visual Display Overlay measurements and bounding circles directly on processed image for clear visualization
o4	o5	
Web Interface Build Flask-based web application allowing easy image upload and results presentation	Computer Vision Pipeline Apply proven techniques including thresholding, morphological operations, and contour analysis	

Technology Stack

Component	Role in System	Key Features
Python	Core programming language	Clean syntax, extensive libraries
OpenCV	Image processing engine	Contour detection, filtering, transformations
NumPy	Mathematical operations	Array manipulation, calculations
Flask	Web framework	Server-side handling, routing
HTML	Frontend interface	User input forms, results display

Each component plays a specialized role in transforming uploaded images into precise measurements.

System Architecture



OpenCV applies detection algorithms



Contour Detection

Find all potential circular objects



Quality Filter

Apply circularity threshold ≥ 0.70



Measurement

Calculate real-world dimensions



Results Display

Show annotated image and data table



Implementation Methodology



Image Preprocessing

- Convert RGB to grayscale channel
- Apply Gaussian blur to reduce noise
- Use adaptive thresholding for binary conversion



Contour Detection

- Find all external contours in binary image
- Filter by minimum area threshold
- Calculate circularity for each contour
- Accept only contours with ≥ 0.70 circularity



Diameter Measurement

- Identify largest detected coin as reference
- Assume reference diameter = 27mm (quarter)
- Calculate pixel-to-mm scaling factor
- Apply scaling to all detected coins

- Morphological operations to close gaps

Mathematical Foundation

Diameter Conversion

$$\frac{\textit{Real Diameter}}{\textit{Coin Pixel Diameter}} = \frac{\textit{Reference Pixel Diameter}}{\textit{Reference Real Diameter}} \times$$

This proportion scales all measurements relative to the known reference coin. For example, if the reference quarter measures 100 pixels (27mm), a coin measuring 80 pixels equals 21.6mm.

Circularity Formula

$$\textit{Circularity} = \frac{4\pi \times \textit{Area}}{(\textit{Perimeter})^2}$$

Circularity values range from 0 to 1. Perfect circles approach 1.0. Applied filter accepts only contours with ≥ 0.70 , ensuring valid coin shapes while rejecting irregular objects.

Results & Technical Advantages

Visual Output

- Original image with annotated circles
- Diameter values overlaid on each coin
- Color-coded bounding regions

Data Table

- Coin ID tracking
- Precise millimeter measurements
- Pixel dimension reference

System Benefits

- Low-cost implementation
- Simple Python-based codebase
- Real-time processing capability

Future Enhancements



Automatic Reference Detection

Implement deep learning classifier to identify coin denominations automatically, eliminating manual reference assumption



Perspective Correction

Add vanishing point detection and image rectification to compensate for angled shots and parallax errors



Deep Learning Pipeline

Replace traditional CV with YOLO or Faster R-CNN architectures for improved detection accuracy and speed



Mobile Application

Build React Native interface with live camera access for instant measurement without image upload step



Extension Possibilities: This technique extends beyond coins to any circular objects. Potential applications include industrial part inspection, botanical measurements, and quality control systems

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